Anna L. Trella

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annatrella.github.io

Education

Department of Computer Science, Harvard University — Cambridge, MA
Ph.D. in Computer Science — 2020 - 2025 *(expected)*
- Advisors: Prof. Susan A. Murphy and Prof. Finale Doshi-Velez
- GPA: 4.00/4.00. Research area: reinforcement learning (RL) and Bayesian machine learning (ML)

Department of Computer Science, Harvard University — Cambridge, MA
M.S. in Computer Science — 2020 - 2024
- GPA: 4.00/4.00. Coursework: reinforcement learning, Bayesian machine learning, interpretable and explainable machine learning, probability theory

Department of Mathematics, University of California, Berkeley — Berkeley, CA
B.A. in Applied Mathematics — 2014 - 2018
- GPA: 3.71/4.00. Coursework: machine learning, artificial intelligence, algorithms & theory, linear algebra, numerical analysis

Experience

Applied Scientist Intern | Amazon Alexa AI — Summer 2023
- Researched and developed a hierarchical reinforcement learning method for fine-tuning large language models (LLMs).
- Tools used: DeepSpeed, Amazon Web Services (AWS), Python

Research Scientist Intern | Etsy — Summer 2021
- Developed and implemented a Bayesian method for accelerating A/B testing using customer e-commerce data
- Tools used: Python, Google Cloud Platform

Technical Lead: Software Development Engineer | Amazon — 2018 - 2020
- Led full-stack development for creating a personalized navigation experience on Whole Foods Market, Prime Now, and Amazon Fresh platforms.
- Tools used: AWS, Scrum, Java, Kotlin, React Native, Objective-C, JavaScript

Skills

Programming: Python, Pytorch, NumPy, Git, DeepSpeed, Tensorflow, PyMC3, Flask, Java, AWS, Xcode, React Native, Objective-C, JavaScript, Kotlin

Languages: English (native), Chinese (native), Spanish (proficient)

Patents

Miguel Roman Durazo Almeida, Thanhthanh Anh Tran, Jongun Jung, Rohin Patel, **Anna L. Trella**, Tarmandeep Toor, Shih-Ta Peng. *Item Identification based on receipt image.* U.S. Patent 11,257,049, filed January 31, 2020, and issued February 22, 2022.

Teaching

Graduate Teaching Fellow, Harvard University
CS 181 Machine Learning (4.95/5.00 on student evaluations) — 2022

Undergraduate Student Instructor, UC Berkeley
CS 61C Machine Structures (4.91/5.00 on student evaluations) — 2017, 2018

Awards

- **Global Symposium on AI in Dentistry Scholarship**, Harvard University — 2023
- **Derek Bok Certificate of Distinction in Teaching**, Harvard University — 2022
- **RLDM Student Travel Fellowship**, Brown University — 2022
- **EECS Outstanding Graduate Student Instructor Award**, UC Berkeley — 2018
- **Dean's Honors List**, UC Berkeley — 2016

Invited Talks

A Deployed Online Reinforcement Learning Algorithm in an Oral Health Clinical Trial
Conference on Statistical Learning and Data Science 2024

Monitoring Fidelity of Reinforcement Learning Algorithms in Clinical Trials
Society for Clinical Trials, Annual Meeting 2024

Oralytics: An Online Reinforcement Learning Algorithm Supporting Oral Self-Care
Harvard University, Guest Lecturer for STAT 234: Sequential Decision-Making 2024

Reward Design for Online Reinforcement Learning Algorithms
Two Sigma Headquarters, PhD Symposium 2023

Designing Reinforcement Learning Algorithms for Digital Interventions
University of Pennsylvania, Ivy Collective Engineering Symposium 2022

Online Reinforcement Learning Algorithms for Mobile Health
mHealthHUB, mDOT Center Webinar Series 2022

Publications

1. **Anna L. Trella**, Kelly W. Zhang, Hinal Jajal, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. "A Deployed Online Reinforcement Learning Algorithm In An Oral Health Clinical Trial." *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI-25)*, 2025.
2. **Anna L. Trella**, Ziming Li, Jinseok Nam, Puyang Xu. "Improving Personalization in Text Generation: A Two-Level Fine-Tuning Approach for LLMs." *In Submission.*
3. **Anna L. Trella**, Walter Dempsey, Ziping Xu, Finale Doshi-Velez, Susan A. Murphy. "Non-Stationary Latent Auto-Regressive Bandits." *In Submission.*
4. **Anna L. Trella**, Susobhan Ghosh, Erin E. Bonar, Lara Coughlin, Finale Doshi-Velez, Yongyi Guo, Pei-Yao Hung, Inbal Nahum-Shani, Vivek Shetty, Maureen Walton, Iris Yan, Kelly W. Zhang, Susan A. Murphy. "Effective Monitoring of Online Decision-Making Algorithms in Digital Intervention Implementation." *In Submission.*
5. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Iris Yan, Finale Doshi-Velez, Susan A. Murphy. "Monitoring Fidelity of Online Reinforcement Learning Algorithms in Clinical Trials." *In Submission.*
6. Kelly W. Zhang, Nowell Closser, **Anna L. Trella**, Susan A. Murphy. "Replicable Bandits for Digital Health Interventions." *In Submission.*
7. Esmaeil Keyvanshokooh, Kyra Gan, Yongyi Guo, Xueqing Liu, **Anna L. Trella**, Susan A. Murphy. "Improving Treatment Responses through Limited Nudges: A Data-Driven Learning and Optimization Approach." *In Submission.*
8. Inbal Nahum-Shani, Zara M. Greer, **Anna L. Trella**, Kelly W. Zhang, Stephanie M. Carpenter, Dennis Ruenger, David Elashoff, Susan A. Murphy, Vivek Shetty. "Optimizing an adaptive digital oral health intervention for promoting oral self-care behaviors: Micro-randomized trial protocol." *Contemporary Clinical Trials*, 2024.
9. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. "Reward Design For An Online Reinforcement Learning Algorithm Supporting Oral Self-Care." *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI-23)*; selected for an oral, 2023.
10. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. "Designing Reinforcement Learning Algorithms for Digital Interventions: Pre-Implementation Guidelines." *Algorithms (Special Issue Algorithms in Decision Support Systems)*, 2022.
    Preliminary version at *RLDM 2022 (Multi-disciplinary Conference on RL and Decision Making)*; selected for an oral (top 7% of accepted papers), 2022.
11. **Anna L. Trella**, Peniel N. Argaw, Michelle M. Li, James A. Hay. "Discrepancies in Epidemiological Modeling of Aggregated Heterogeneous Data." *IJCAI 2021 Workshop on Artificial Intelligence for Social Good*; selected for an oral, 2021.

Academic Service

Reviewer: *Reinforcement Learning Conference (RLC) 2024*
Innovative Applications of Artificial Intelligence (IAAI) 2024

Hobbies

Figure Skating (Senior Moves in the Field Gold Medalist)
Cheerleading (NCA National Champion, 2x California State Champion)
Piano (Certificate of Merit Level 6)